



LR-N22-0020
March 18, 2022

10CFR50.73

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington DC 20555-001

Hope Creek Generating Station
Renewed Facility Operating License No. NPF-57
Docket No. 50-354

Subject: Licensee Event Report 2022-001-00
B EDG Inoperable Resulting in a Condition Prohibited by Technical
Specifications

In accordance with 10 CFR 50.73(a)(2)(i)(B), PSEG Nuclear is submitting Licensee Event Report (LER) number 2022-001-00, "B EDG Inoperable Resulting in a Condition Prohibited by Technical Specifications."

There are no regulatory commitments contained in this letter.

If you have any questions or require additional information, please contact Mr. Justin Wearne.

Sincerely,

A handwritten signature in black ink, appearing to read "SR Poorman", written over a horizontal line.

Steven R. Poorman
Plant Manager
Hope Creek Generating Station

Attachment: Licensee Event Report 2022-001-00

cc: Regional Administrator – USNRC Region I
USNRC Project Manager – Hope Creek
USNRC Senior Resident Inspector – Hope Creek (X-24)
Manager IV, NJ Bureau of Nuclear Engineering
Corporate Commitment Coordinator



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk a/c: omb_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Hope Creek Generating Station

2. Docket Number

05000

00354

3. Page

1 OF 3

4. Title

B EDG Inoperable Resulting in a Condition Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
01	19	2022	22	- 001 -	00	03	18	2022	Facility Name	Docket Number
										05000
									Facility Name	Docket Number
										05000

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☐ 10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	☐ 10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	☐ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	
☐ OTHER (Specify here, in abstract, or NRC 366A).				

12. Licensee Contact for this LER

Licensee Contact

Justin Wearne, Manager – Site Compliance

Phone Number (Include area code)

856-339-1351

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected



No



Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month

Day

Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On January 19, 2022 during the performance of the B Emergency Diesel Generator (EDG) 24-hour surveillance test, the B EDG did not achieve the required load of 4652 kW-4873 kW (105 – 110 percent of rated load). The maximum load achieved was 4650 kW (104.9 percent). The B EDG was determined to be inoperable due to failing to meet the acceptance criteria. Investigation identified that the B EDG fuel supply linkage mechanism (fuel rack) required adjustment. Following adjustment, the B EDG was re-tested satisfactorily. The B EDG was returned to operable status on January 23, 2022 at 0700.

Technical Specification (TS) 3.8.1.1 allows the B EDG to remain inoperable for up to 14 days. Based on a review of the B EDG maintenance history, it was determined that the condition existed prior to discovery and for a period greater than 14 days. Therefore, an Operation or Condition Prohibited by Technical Specifications occurred, and this report is being made per the requirements of 10CFR50.73(a)(2)(i)(B).

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME		2. DOCKET NUMBER		3. LER NUMBER		
Hope Creek Generating Station		05000-	00354	YEAR 22	SEQUENTIAL NUMBER 001	REV NO. 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric - Boiling Water Reactor (BWR/4)
Emergency Onsite Power Supply System - EIS Identifier {EK}
Diesel Generator - EIS Identifier {DG}

IDENTIFICATION OF OCCURRENCE

Event Date: May 3, 2021
Discovery Date: January 19, 2022

CONDITIONS PRIOR TO OCCURRENCE

Hope Creek was in Operational Condition (OPCON) 1, at 100 percent rated thermal power. No other structures, systems or components that could have contributed to the event were inoperable at the time of the event.

DESCRIPTION OF OCCURRENCE

On January 19, 2022 Hope Creek was performing TS surveillance requirement 4.8.1.1.2.k, which requires that the B Emergency Diesel Generator (EDG) be operated for 22 hours at rated load, followed by 2 hours at a load of 4652 kW - 4873 kW (105 to 110 percent of rated load). This test is performed once per 20.5 months. Following successful operation at rated load for 22 hours, the maximum load that could be achieved was 4650 kW for the final portion of the surveillance test. Based on this condition, the B EDG was determined to be inoperable on January 19, 2022 at 12:59.

An investigation determined that the B EDG cylinder injection pumps needed adjustment to increase fueling to the engine. This is performed by adjusting a mechanical linkage (fuel rack). Following adjustment, the EDG was re-tested successfully and declared operable.

CAUSE

An investigation of the maintenance history determined that the inability of the B EDG to reach max load condition was introduced when fuel rack adjustments were made in April and May, 2021 during planned maintenance conducted during the Hope Creek refuel outage 23. Following the adjustments, the EDG was tested to 4000-4400 kW per TS surveillance requirement 4.8.1.1.2.a, on May 3, 2021. It was not tested at the higher load of 105-110 percent. The condition on the B EDG existed from May 3, 2021 until resolved on January 23, 2022. This exceeds the TS allowed outage time of 14 days.

SAFETY CONSEQUENCES

Per the Hope Creek Updated Final Safety Analysis Report (UFSAR) and supporting calculations, the maximum EDG loading for any of the EDGs under accident conditions is 4628.44 kW. Under the condition described above, the B EDG was able to be loaded to 4650 kW, therefore the B EDG was able to satisfy the design basis loading requirements.

PREVIOUS EVENTS

A review of station Licensee Event Reports and the corrective action program for the past 3 years was conducted. No LERs were identified. One corrective action occurrence was identified. In October, 2019 the C EDG would not achieve more than 4700 kW following a planned governor replacement. The condition was identified during post maintenance testing and was corrected prior to returning the EDG to operable status.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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Hope Creek Generating Station	05000- 00354	YEAR 22	- SEQUENTIAL NUMBER 001	- REV NO. 00

NARRATIVE**CORRECTIVE ACTIONS**

On January 22, 2022, the B EDG fuel rack mechanical linkage was adjusted to allow the required loading of 4652 kW - 4873 kW to be achieved.

The remaining three EDGs (A, C and D) were verified to have successfully completed TS surveillance requirement 4.8.1.1.2.k (105 - 110 percent loaded run) subsequent to refueling outage 23. A review of maintenance activities found no history of fuel rack adjustments since the last successful completion of surveillance requirement 4.8.1.1.2.k.

A cause evaluation is in progress to identify additional actions.

COMMITMENTS

None